

ROJALEEN SAHU

Electronics & Communication Engineer | Software Development | Data Analytics | IoT

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PROFESSIONAL SUMMARY

Motivated Electronics and Communication Engineering graduate (BPUT, 2026) with strong programming foundations in Java, Python, C, and C++. Hands-on experience in IoT systems, SQL databases, and data analytics using Power BI. Exposure to SAP enterprise tools (ABAP, Fiori, IBP) and real-world industrial automation at Rourkela Steel Plant. Built software-driven projects including a Digital Attendance System, Smart Energy Meter, and an AI-powered Heart Disease Risk Prediction system. Passionate about software development, data-driven problem solving, and leveraging technology to create impactful solutions — aligned with Infosys's AI-first digital vision.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++

Database & Query Language: SQL (MySQL) — Queries, Joins, Aggregations

Data Analytics & BI: Power BI, AI-Driven Data Analysis, Microsoft Excel (Advanced)

Web & Enterprise Tools: SAP Back-End ABAP, SAP Fiori, SAP IBP

IoT & Embedded Systems: Arduino IDE, NodeMCU, Sensor Integration, Cloud IoT Platforms

Simulation & Modelling: MATLAB

Productivity Suite: Microsoft Word, Excel, PowerPoint

INTERNSHIP EXPERIENCE

IoT Intern | OCAC, Bhubaneswar

May 2025 – July 2025

- Designed and deployed smart IoT systems — home automation and automated irrigation — using NodeMCU and Arduino, applying core programming logic in C/C++.
- Integrated multiple sensors for real-time data acquisition and monitoring via cloud-based IoT platforms, strengthening hardware-software interface skills.
- Streamlined hardware-software communication pipelines, improving overall system efficiency and reliability for real-world automation challenges.

Electronics & Instrumentation Intern | RSP – Rourkela Steel Plant

May 2024

- Gained hands-on exposure to large-scale industrial automation systems and sophisticated control systems within a manufacturing environment.
- Assisted in maintenance and troubleshooting of process control instrumentation used in high-precision steel production workflows.
- Observed end-to-end operational workflows, strengthening understanding of system-level engineering and process efficiency.

PROJECTS

Intelligent Heart Disease Risk Prediction System | Academic Project

2025

- Developed an AI-powered heart disease risk prediction system using Python and machine learning algorithms (Logistic Regression, Random Forest, XGBoost) trained on clinical datasets.

- Performed data preprocessing and feature engineering on patient health parameters including age, cholesterol, blood pressure, ECG results, and lifestyle indicators.
- Achieved high prediction accuracy through model tuning, cross-validation, and hyperparameter optimization, enabling reliable early-risk identification.
- Built an interactive dashboard using Power BI / Streamlit to visualize risk scores, feature importance, and patient-level insights for clinical decision support.
- Demonstrated practical application of AI in healthcare, contributing to early intervention strategies and reduced diagnostic burden.

Student Digital Attendance System | Academic Project

2024 – 2025

- Developed a fully automated digital attendance management system using Python and SQL backend to track and store student records efficiently.
- Implemented RFID/biometric sensor integration for real-time attendance capture, eliminating manual data entry errors.
- Designed interactive Power BI dashboards for attendance trend analysis, enabling data-driven insights for faculty and administrators.
- Reduced administrative effort by ~80% compared to traditional paper-based systems, demonstrating impact through measurable outcomes.
- Applied object-oriented programming (OOP) concepts in Python to build a modular, scalable codebase.

Digital Smart Energy Meter | Academic Project

2024

- Designed and developed a smart prepaid energy meter with billing management system for efficient energy consumption tracking.
- Integrated over-voltage and under-voltage protection logic using embedded C programming for improved system safety.
- Implemented real-time electricity theft detection algorithm to identify and flag unauthorized power usage automatically.

EDUCATION

B.Tech – Electronics & Communication Engineering

Biju Patnaik University of Technology (BPUT), Odisha | Expected: June 2026

- CGPA: 7.6 (Simple Average across all semesters including labs and electives)
- Key Coursework: Data Structures, Digital Electronics, Digital Signal Processing, Microcontrollers, Operating Systems, Computer Networks.
- Applied theoretical concepts through hands-on lab projects in microcontroller programming, circuit design, and embedded systems.